

Advanced Python Threading

A complete, in-depth guide to concurrency with the threading module — internals, synchronization primitives, patterns, pitfalls, and production practice.

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1. The GIL and Why Threading Works the Way It Does

The Global Interpreter Lock (GIL) is a mutex inside CPython (the reference implementation most people mean when they say "Python") that allows only **one thread to execute Python bytecode at a time**, even on a multi-core machine. Understanding it is the single most important prerequisite to using threading correctly, because it determines what threads can and cannot speed up.

1.1 Why the GIL exists

CPython manages memory with reference counting. Every object has a counter that is incremented and decremented as references are created and destroyed; when it hits zero the object is freed. Without a global lock, two threads could increment/decrement the same counter simultaneously, corrupting it (a classic race condition) and causing memory leaks or premature frees / crashes. The GIL was the simplest way to make the reference-counting memory manager thread-safe without redesigning it, at the cost of true parallel bytecode execution.

1.2 What the GIL means in practice

- **CPU-bound code** (tight Python loops, number crunching in pure Python) does not get faster with more threads — **the GIL serializes bytecode execution**, so 4 threads doing pure math run at roughly the same wall-clock speed as 1 thread, sometimes slower due to context-switch overhead.
- **I/O-bound code** (network calls, disk reads, database queries, `time.sleep`) benefits enormously from threads. When a thread performs a blocking I/O call, it releases the GIL, allowing another thread to run while the first waits for the OS/kernel/network.
- C extensions that do heavy numeric work (NumPy, hashlib, zlib, most C-accelerated libraries) release the GIL internally during their computation, so threads calling into them CAN achieve real parallelism even though it's "Python" threading.

```
import time
import threading

def cpu_bound(n):
    x = 0
    for i in range(n):
        x += i * i
    return x

N = 40_000_000

# Single-threaded baseline
start = time.perf_counter()
cpu_bound(N)
cpu_bound(N)
print("Sequential:", time.perf_counter() - start)

# Two threads sharing the work
start = time.perf_counter()
t1 = threading.Thread(target=cpu_bound, args=(N,))
t2 = threading.Thread(target=cpu_bound, args=(N,))
t1.start(); t2.start()
t1.join(); t2.join()
```

```
print("Two threads:", time.perf_counter() - start)
```

```
Sequential: 3.41s
```

```
Two threads: 3.47s <-- NOT faster: the GIL serialized both threads
```

Note: Python 3.13 introduced an experimental **free-threaded build** (PEP 703, python3.13t) that can disable the GIL entirely. As of this writing it is still opt-in and the ecosystem (C extensions) is still catching up, so the concepts in this guide — locks, races, deadlocks — remain essential even in that build; in fact they become MORE important because CPU-bound races that the GIL used to mask become real races.

1.3 The rule of thumb

Use **threading** for I/O-bound concurrency (web scraping, API calls, file/network I/O, database queries). Use **multiprocessing** or a C-accelerated library for CPU-bound parallelism. This single decision saves more debugging time than any other in this guide.

2. Creating and Managing Threads

2.1 The Thread class — function based

The most common way to create a thread is instantiating `threading.Thread` with a target callable and arguments. The thread does not start running until `.start()` is called.

```
import threading
import time

def worker(name, delay):
    print(f"[{name}] starting")
    time.sleep(delay)
    print(f"[{name}] finished")

t = threading.Thread(target=worker, args=("Worker-1",), kwargs={"delay": 1.5})
t.start()
print("Main thread continues immediately")
t.join() # block until t finishes
print("Main thread done")
```

2.2 Subclassing Thread

For more complex, stateful workers, subclass `Thread` and override `run()`. This is preferred when the thread needs its own **internal state, multiple methods, or lifecycle hooks**.

```
import threading
import time

class DownloaderThread(threading.Thread):
    def __init__(self, url, results):
        super().__init__(name=f"Downloader-{url}")
        self.url = url
        self.results = results # shared dict, must be protected if written concurrently

    def run(self):
        # self.run() is invoked automatically by .start(); never call run() directly
        # or it executes synchronously in the calling thread instead of a new one.
        time.sleep(0.5) # simulate network I/O
        self.results[self.url] = f"200 OK ({self.url})"

results = {}
threads = [DownloaderThread(url, results) for url in
            ["site-a.com", "site-b.com", "site-c.com"]]

for t in threads:
    t.start()
for t in threads:
    t.join()

print(results)
```

Pitfall: Never call `thread.run()` directly — that executes the method synchronously on the current thread, defeating the purpose. Always call `thread.start()`.

2.3 Passing arguments safely

Arguments passed via `args`/`kwargs` are bound at thread-creation time, which avoids the classic "closure captures the loop variable by reference" bug that trips people up when starting threads in a loop.

```
import threading

# BUGGY: all threads may print the same (last) value of i due to late binding
for i in range(3):
    threading.Thread(target=lambda: print(i)).start()

# CORRECT: bind i as a default argument, evaluated immediately
for i in range(3):
    threading.Thread(target=lambda i=i: print(i)).start()

# CORRECT (idiomatic): pass via args
def show(i):
    print(i)
for i in range(3):
    threading.Thread(target=show, args=(i,)).start()
```

2.4 Naming threads for debuggability

Give threads meaningful names via `name=`; it shows up in tracebacks, logs, and tools like `threading.enumerate()`, which is invaluable once you have more than two or three threads running.

```
import threading, logging

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(threadName)s] %(message)s"
)

def task():
    logging.info("doing work")

threading.Thread(target=task, name="ImageResizer-1").start()
threading.Thread(target=task, name="ImageResizer-2").start()
```

3. Thread Lifecycle, Daemon Threads & Joining

3.1 Lifecycle states

A thread moves through: **created** (object instantiated) → **started** (`.start()` called, OS thread spawned, `run()` begins executing) → **running/blocked** (executing, or waiting on I/O / a lock / a condition) → **terminated** (`run()` returns or raises). You can query state with `t.is_alive()`; **there is no way to forcibly kill a thread in Python — it must exit cooperatively.**

3.2 `join()` and timeouts

`join()` blocks the calling thread until the target thread finishes. Passing a timeout lets you wait bounded time and check `is_alive()` afterwards to know whether it actually finished or you just gave up waiting.

```
import threading, time

def slow_task():
    time.sleep(3)

t = threading.Thread(target=slow_task)
t.start()
t.join(timeout=1.0)

if t.is_alive():
    print("Still running after 1s – did not finish in time, thread was NOT killed")
else:
    print("Finished within 1s")
```

3.3 Daemon threads

A **daemon thread** is a background thread that does not prevent the interpreter from exiting. **When the main thread finishes and only daemon threads remain, Python exits immediately, abandoning them mid-execution (no cleanup code, no `finally` blocks guaranteed to run).** Non-daemon ("normal") threads, by contrast, **keep the process alive until they finish.**

```
import threading, time

def heartbeat():
    while True:
        print("...still alive...")
        time.sleep(1)

t = threading.Thread(target=heartbeat, daemon=True) # must set before start()
t.start()

time.sleep(3.5)
print("Main thread exiting now")
# The heartbeat thread is killed abruptly here – no cleanup runs.
```

- Use daemon threads for background maintenance work you're fine abandoning on shutdown (e.g. a metrics pinger, a cache-warming loop).
- Use non-daemon threads (the default) for work that **MUST** complete — writing a file, flushing a database transaction, sending a critical network request.

- `daemon=True` must be set before `start()`; changing it afterward raises `RuntimeError`.

Pitfall: Daemon threads are not a substitute for graceful shutdown. For anything with side effects (open files, database connections, partially written data) use an explicit stop mechanism — typically a `threading.Event` the loop checks each iteration — rather than relying on abrupt daemon termination.

```
import threading, time

stop_event = threading.Event()

def graceful_worker():
    while not stop_event.is_set():
        print("working...")
        time.sleep(0.5)
    print("cleaning up and exiting")

t = threading.Thread(target=graceful_worker)
t.start()
time.sleep(2)
stop_event.set()      # ask the thread to stop
t.join()              # wait for it to actually finish and clean up
```

4. Race Conditions and the Lock Primitive

4.1 Anatomy of a race condition

A race condition occurs when **multiple threads read and write shared state without coordination**, and the final result depends on the **unpredictable timing of context switches**. The classic example is an unprotected counter increment, which is NOT atomic in Python even though it looks like one line — `counter += 1` compiles to load-add-store bytecode, and a thread can be interrupted between those steps.

```
import threading

counter = 0

def increment():
    global counter
    for _ in range(200_000):
        counter += 1      # NOT atomic: read, add 1, write back

threads = [threading.Thread(target=increment) for _ in range(4)]
for t in threads: t.start()
for t in threads: t.join()

print(counter)    # Expected 800000, but usually prints something LESS, e.g. 734158
```

This happens even with the GIL, because the GIL **only guarantees that a single bytecode instruction executes atomically** — not a whole Python statement, and definitely not a multi-instruction sequence like a compound assignment.

4.2 Fixing it with Lock

`threading.Lock` is a mutex: `acquire()` blocks until the lock is free then claims it; `release()` frees it. **Only one thread can hold a given Lock at a time**, so wrapping the critical section makes the compound operation effectively atomic with respect to other threads using the same lock.

```
import threading

counter = 0
lock = threading.Lock()

def increment():
    global counter
    for _ in range(200_000):
        with lock:          # equivalent to lock.acquire() ... lock.release()
            counter += 1

threads = [threading.Thread(target=increment) for _ in range(4)]
for t in threads: t.start()
for t in threads: t.join()

print(counter)    # Reliably 800000
```

Note: **Always prefer the `with lock: context-manager` form over manual `acquire()/release()`.** It guarantees the lock is released even if an exception is raised inside the block — manual release calls are easy to skip on an exception path, which causes a permanent deadlock.

4.3 Lock granularity and performance

Holding a lock across expensive work (I/O, sleeping, heavy computation) serializes threads and defeats the purpose of concurrency. Keep critical sections as small as possible: acquire the lock only around the shared-state mutation, not around unrelated work.

BAD: lock held during slow I/O, other threads pile up waiting for no reason

```
def bad_worker(shared, lock):
    with lock:
        data = fetch_from_network() # slow, doesn't need the lock
        shared.append(data)
```

GOOD: only the shared-state mutation is protected

```
def good_worker(shared, lock):
    data = fetch_from_network() # runs concurrently across threads
    with lock:
        shared.append(data) # brief critical section
```

4.4 acquire() with timeout and non-blocking tries

```
import threading

lock = threading.Lock()

if lock.acquire(timeout=2):
    try:
        pass # got the lock within 2 seconds
    finally:
        lock.release()
else:
    print("Could not acquire lock in time – back off or retry")

# Non-blocking attempt
if lock.acquire(blocking=False):
    try:
        pass
    finally:
        lock.release()
else:
    print("Lock busy, skip this cycle")
```

5. RLock (Reentrant Lock)

A plain Lock cannot be acquired twice by the same thread — doing so deadlocks the thread against itself. `threading.RLock` (reentrant lock) tracks WHICH thread owns it and how many times it has been acquired, allowing the owning thread to re-acquire it without blocking. It must be released the same number of times it was acquired before another thread can take it.

5.1 Why you need it: recursive or nested locking

```
import threading

class Account:
    def __init__(self, balance):
        self.balance = balance
        self.lock = threading.RLock()  # RLock, not Lock

    def deposit(self, amount):
        with self.lock:
            self.balance += amount

    def transfer_in_and_log(self, amount):
        with self.lock:
            # first acquire
            self.deposit(amount)  # deposit() acquires again -> fine with RLock
            print(f"New balance: {self.balance}")

acct = Account(100)
acct.transfer_in_and_log(50)
```

Pitfall: If `Account.lock` above were a plain `Lock` instead of an `RLock`, calling `transfer_in_and_log` would deadlock: the thread would block forever trying to re-acquire a lock it already holds.

5.2 When to use RLock vs Lock

- Use **RLock** when a locked method may **call another locked method** on the same object from the same thread (common in class-based designs with public methods that call each other internally).
- Use plain **Lock** everywhere else — it's slightly cheaper and its stricter behavior helps catch accidental re-entrancy bugs during development.
- **RLock** does NOT make an object safe for use by other threads while held; it only allows the SAME thread to nest acquisitions. Other threads still block until all nested releases complete.

6. Semaphore and BoundedSemaphore

A Semaphore maintains an internal counter. `acquire()` decrements it (blocking if it would go below zero), and `release()` increments it. Unlike a Lock (capacity 1), a Semaphore with counter N allows up to N threads into the critical section concurrently — **it's the tool for limiting concurrency, not for mutual exclusion**.

6.1 Limiting concurrent access — e.g. rate-limiting API calls

```
import threading, time

MAX_CONCURRENT_REQUESTS = 3
semaphore = threading.Semaphore(MAX_CONCURRENT_REQUESTS)

def call_api(i):
    with semaphore:
        print(f"[{i}] calling API ({threading.active_count()-1} threads total)")
        time.sleep(1)    # simulate the request
        print(f"[{i}] done")

threads = [threading.Thread(target=call_api, args=(i,)) for i in range(10)]
for t in threads: t.start()
for t in threads: t.join()
# At most 3 threads are ever inside the "with semaphore" block at once.
```

6.2 BoundedSemaphore

BoundedSemaphore behaves like Semaphore but raises `ValueError` if `release()` is called more times than `acquire()`, catching a common bug (an extra release, e.g. from a duplicated `finally` block) that a plain Semaphore would silently allow, corrupting your concurrency limit.

```
import threading

sem = threading.BoundedSemaphore(2)
sem.acquire()
sem.release()
sem.release()    # raises ValueError: Semaphore released too many times
```

Note: Practical rule: default to **BoundedSemaphore** in application code — the extra safety check almost never costs anything and catches real bugs.

7. Event

An Event is the simplest thread-signaling primitive: an internal boolean flag, initially False. Threads call `wait()` to block until the flag becomes True; any thread can flip it with `set()` (waking every waiter) or reset it with `clear()`. Use it for one-to-many "something happened" signals — it does not carry data and does not enforce mutual exclusion.

7.1 Coordinating startup / shutdown

```
import threading, time

ready = threading.Event()
shutdown = threading.Event()

def worker(worker_id):
    print(f"Worker {worker_id} waiting for ready signal...")
    ready.wait()          # blocks here until main thread calls ready.set()
    print(f"Worker {worker_id} started")
    while not shutdown.is_set():
        time.sleep(0.3)
        # do periodic work
    print(f"Worker {worker_id} shutting down cleanly")

threads = [threading.Thread(target=worker, args=(i,)) for i in range(3)]
for t in threads: t.start()

time.sleep(1)
print("Main: releasing workers")
ready.set()          # all 3 workers unblock at once

time.sleep(1.5)
print("Main: signaling shutdown")
shutdown.set()
for t in threads: t.join()
```

7.2 wait() with timeout

```
signal = threading.Event()

if signal.wait(timeout=5):
    print("Signal received within 5 seconds")
else:
    print("Timed out waiting for signal")
```

8. Condition Variables

A Condition pairs a Lock with the ability to `wait()` for a signal and `notify()` waiters, atomically releasing the lock while waiting and reacquiring it upon wake-up. It is the right tool when threads need to wait for an arbitrary, checkable condition on shared state ("wait until the queue is non-empty", "wait until capacity is available") rather than a one-shot flag.

8.1 The wait-notify pattern

Pitfall: Always call `wait()` inside a while loop that re-checks the condition, never a plain if. "Spurious wakeups" and multiple waiters racing to consume one notification mean the condition might no longer hold by the time your thread actually resumes.

```
import threading, collections, time, random

condition = threading.Condition()
buffer = collections.deque(maxlen=5)

def producer():
    for i in range(10):
        time.sleep(random.uniform(0.05, 0.2))
        with condition:
            while len(buffer) == buffer.maxlen:
                condition.wait()          # buffer full, wait for a consumer to make room
            buffer.append(i)
            print(f"Produced {i}, buffer={list(buffer)}")
            condition.notify()            # wake one waiting consumer

def consumer():
    for _ in range(10):
        with condition:
            while not buffer:
                condition.wait()          # buffer empty, wait for a producer
            item = buffer.popleft()
            print(f"Consumed {item}")
            condition.notify()            # wake one waiting producer

t1 = threading.Thread(target=producer)
t2 = threading.Thread(target=consumer)
t1.start(); t2.start()
t1.join(); t2.join()
```

8.2 `notify()` vs `notify_all()`

- `notify(n=1)` wakes only the given number of waiting threads — use it when you know exactly how many can proceed (e.g. exactly one slot freed up).
- `notify_all()` wakes every waiter — safer default when you're not sure how many can proceed, at the cost of extra wake-ups that immediately re-check and re-sleep if their condition still isn't met.

9. Barrier

A Barrier makes a fixed number of threads wait for each other at a synchronization point: no thread proceeds past `barrier.wait()` until ALL parties have called it. It's ideal for phased/lock-step computations where every thread must finish phase N before any thread starts phase N+1.

```
import threading, time, random

NUM_WORKERS = 4
barrier = threading.Barrier(NUM_WORKERS)

def worker(worker_id):
    for phase in range(3):
        duration = random.uniform(0.1, 0.6)
        time.sleep(duration)
        print(f"Worker {worker_id} finished phase {phase}")
        barrier.wait()      # blocks until all 4 workers reach this point
        if worker_id == 0:
            print(f"--- All workers completed phase {phase} ---")

threads = [threading.Thread(target=worker, args=(i,)) for i in range(NUM_WORKERS)]
for t in threads: t.start()
for t in threads: t.join()
```

Note: If a thread never reaches the barrier (crashes, times out) the others waiting there will raise `BrokenBarrierError` rather than hang forever — pass `timeout=` to `wait()` to bound how long a thread waits before that happens.

10. Thread-Safe Queues & Producer-Consumer

`queue.Queue` is a FIFO queue with all locking built in — it is by far the most common and safest way to move data between threads, and usually replaces manual Condition-based producer/consumer code entirely. The module also provides `LifoQueue` (stack order) and `PriorityQueue`.

10.1 Basic producer-consumer with Queue

```
import threading, queue, time

q = queue.Queue(maxsize=10)  # bounded queue applies natural backpressure

def producer():
    for i in range(20):
        q.put(i)                # blocks if the queue is full
        print(f"Produced {i}")
    q.put(None)                 # sentinel value to signal "no more items"

def consumer():
    while True:
        item = q.get()          # blocks until an item is available
        if item is None:
            q.task_done()
            break
        print(f"Consumed {item}")
        time.sleep(0.05)
        q.task_done()           # signals this item's processing is complete

t1 = threading.Thread(target=producer)
t2 = threading.Thread(target=consumer)
t1.start(); t2.start()
t1.join(); t2.join()
```

10.2 Multiple consumers and join()

`Queue.join()` blocks until every item put into the queue has been matched by a `task_done()` call — a clean way to wait for a whole batch of work across many worker threads without tracking each one individually.

```
import threading, queue

q = queue.Queue()
for job in range(30):
    q.put(job)

def worker():
    while True:
        job = q.get()
        try:
            process(job)        # your actual work
        finally:
            q.task_done()       # ALWAYS call this, even on exception

def process(job):
    pass  # ... real work here
```

```
NUM_WORKERS = 5
workers = [threading.Thread(target=worker, daemon=True) for _ in range(NUM_WORKERS)]
for w in workers:
    w.start()

q.join() # blocks here until all 30 jobs have been task_done()'d
print("All jobs processed")
```

Pitfall: Forgetting `task_done()` — especially on an exception path — makes `q.join()` hang forever. Put it in a *finally block*.

10.3 Non-blocking and timed operations

```
import queue

q = queue.Queue()

try:
    item = q.get_nowait()
except queue.Empty:
    print("Nothing to process right now")

try:
    q.put_nowait("data")
except queue.Full:
    print("Queue is full, apply backpressure or drop")

try:
    item = q.get(timeout=2)
except queue.Empty:
    print("Nothing arrived within 2 seconds")
```

11. Thread-Local Storage

`threading.local()` creates an object whose attributes are independent per thread — each thread sees its own private copy, with no locking needed and no risk of one thread's writes leaking into another's view. It's the standard way to give each thread its own "global-looking" state, such as a database connection, a request context, or a per-thread buffer.

```
import threading

_local = threading.local()

def get_connection():
    if not hasattr(_local, "conn"):
        _local.conn = create_expensive_db_connection() # once per thread
        print(f"{threading.current_thread().name}: created new connection")
    return _local.conn

def create_expensive_db_connection():
    return object() # placeholder

def handle_request():
    conn = get_connection() # first call per thread creates it, later calls reuse it
    # ... use conn ...

threads = [threading.Thread(target=handle_request, name=f"Handler-{i}") for i in range(3)]
for t in threads: t.start()
for t in threads: t.join()

Handler-0: created new connection
Handler-1: created new connection
Handler-2: created new connection
```

Note: This is exactly the mechanism connection-pool libraries (SQLAlchemy's `scoped_session`, many DB drivers) use internally to hand each thread its own connection safely without you writing any explicit locking.

11.1 Subclassing `threading.local` for per-thread initialization

```
import threading, random

class PerThreadState(threading.local):
    def __init__(self):
        # __init__ runs once PER THREAD the first time that thread touches an attribute
        self.request_id = random.randint(1000, 9999)

state = PerThreadState()

def show():
    print(threading.current_thread().name, state.request_id)

for i in range(3):
    threading.Thread(target=show, name=f"T{i}").start()
```

12. Timer Threads

`threading.Timer` is a `Thread` subclass that runs its target once after a delay. It's useful for one-shot deferred actions (timeouts, retry scheduling) but is NOT a general-purpose scheduler — for repeating jobs, either re-arm a new `Timer` at the end of the callback or use a dedicated scheduling library (e.g. `APScheduler`).

```
import threading

def timeout_handler():
    print("Operation timed out!")

timer = threading.Timer(5.0, timeout_handler)
timer.start()

# If the real operation finishes first, cancel the timer:
# timer.cancel()
```

12.1 Self-repeating timer pattern

```
import threading

class RepeatingTimer:
    def __init__(self, interval, function, *args, **kwargs):
        self.interval = interval
        self.function = function
        self.args = args
        self.kwargs = kwargs
        self._timer = None
        self._stopped = threading.Event()

    def _run(self):
        if not self._stopped.is_set():
            self.function(*self.args, **self.kwargs)
            self._schedule()

    def _schedule(self):
        self._timer = threading.Timer(self.interval, self._run)
        self._timer.daemon = True
        self._timer.start()

    def start(self):
        self._schedule()

    def stop(self):
        self._stopped.set()
        if self._timer:
            self._timer.cancel()

rt = RepeatingTimer(2.0, lambda: print("tick"))
rt.start()
# later: rt.stop()
```

13. ThreadPoolExecutor & concurrent.futures

Manually managing a pool of Thread objects (spawning, joining, tracking results, catching exceptions) is repetitive and error-prone. `concurrent.futures.ThreadPoolExecutor` handles this: a fixed pool of worker threads pulls tasks from an internal queue and hands back Future objects representing eventual results. This is the recommended high-level API for almost all threaded work in modern Python.

13.1 submit() and Future objects

```
from concurrent.futures import ThreadPoolExecutor
import time

def fetch(url):
    time.sleep(1)
    return f"content of {url}"

urls = ["a.com", "b.com", "c.com", "d.com"]

with ThreadPoolExecutor(max_workers=3) as executor:
    futures = [executor.submit(fetch, url) for url in urls]
    for future in futures:
        print(future.result())  # blocks until that specific future completes
# executor.shutdown(wait=True) is called automatically by the "with" block
```

13.2 map() for simple fan-out

```
from concurrent.futures import ThreadPoolExecutor
import time

def fetch(url):
    time.sleep(1)
    return f"content of {url}"

urls = ["a.com", "b.com", "c.com", "d.com"]

with ThreadPoolExecutor(max_workers=4) as executor:
    for result in executor.map(fetch, urls):  # preserves input order
        print(result)
```

13.3 Choosing max_workers

- For I/O-bound work, workers can safely exceed the CPU core count — a common starting point is `min(32, os.cpu_count() + 4)` (the executor's own default), but the right number ultimately depends on how much the remote service/API/DB can handle concurrently.
- Too many workers against a single downstream service can overwhelm it or hit rate limits — pair the pool with a Semaphore or a lower `max_workers` if needed.
- For CPU-bound work, prefer `ProcessPoolExecutor` (same API, separate processes, sidesteps the GIL) instead of `ThreadPoolExecutor`.

14. Futures Deep Dive (as_completed, wait, exceptions)

14.1 as_completed — process results as they finish, not in submission order

```
from concurrent.futures import ThreadPoolExecutor, as_completed
import random, time

def fetch(name):
    time.sleep(random.uniform(0.5, 2.5))
    return name

names = ["fast", "medium", "slow", "very-slow"]

with ThreadPoolExecutor(max_workers=4) as executor:
    future_to_name = {executor.submit(fetch, n): n for n in names}
    for future in as_completed(future_to_name):
        name = future_to_name[future]
        print(f"{name} finished first-to-complete order: {future.result()}")
```

14.2 Exception propagation

If the callable raises, the exception is captured on the Future rather than crashing the worker thread; it is re-raised when you call `.result()`. Forgetting to call `.result()` (or check `.exception()`) means a failure can silently vanish.

```
from concurrent.futures import ThreadPoolExecutor

def risky(x):
    if x == 3:
        raise ValueError(f"bad value: {x}")
    return x * 2

with ThreadPoolExecutor(max_workers=4) as executor:
    futures = [executor.submit(risky, i) for i in range(5)]
    for f in futures:
        try:
            print(f.result())
        except ValueError as e:
            print(f"Task failed: {e}")
```

14.3 wait() with FIRST_COMPLETED / ALL_COMPLETED

```
from concurrent.futures import ThreadPoolExecutor, wait, FIRST_COMPLETED
import time

def task(n):
    time.sleep(n)
    return n

with ThreadPoolExecutor(max_workers=3) as executor:
    futures = {executor.submit(task, t) for t in [1, 2, 3]}
    done, not_done = wait(futures, timeout=1.5, return_when=FIRST_COMPLETED)
    print("Done so far:", [f.result() for f in done])
    print("Still running:", len(not_done))
```

14.4 Cancelling pending (not-yet-started) work

```
from concurrent.futures import ThreadPoolExecutor
import time

with ThreadPoolExecutor(max_workers=1) as executor:
    f1 = executor.submit(time.sleep, 3)    # occupies the only worker
    f2 = executor.submit(time.sleep, 3)    # queued, not yet started
    cancelled = f2.cancel()                # True: it hadn't started yet
    print("Cancelled:", cancelled)
    # Note: cancel() only works BEFORE a task starts running;
    # once running, there is no built-in way to forcibly stop a thread.
```

15. Deadlocks, Livelocks & Starvation

15.1 Classic deadlock: inconsistent lock ordering

The most common deadlock happens when two threads acquire two locks in opposite order. Thread A holds Lock 1 and waits for Lock 2, while Thread B holds Lock 2 and waits for Lock 1 — neither can proceed.

```
import threading, time

lock_a = threading.Lock()
lock_b = threading.Lock()

def thread_1():
    with lock_a:
        time.sleep(0.1)          # gives thread_2 time to grab lock_b
        with lock_b:             # DEADLOCK: waits forever for lock_b
            print("Thread 1 got both locks")

def thread_2():
    with lock_b:
        time.sleep(0.1)          # DEADLOCK: waits forever for lock_a
        with lock_a:
            print("Thread 2 got both locks")

t1 = threading.Thread(target=thread_1)
t2 = threading.Thread(target=thread_2)
t1.start(); t2.start()
t1.join(); t2.join()    # this program hangs forever
```

15.2 Fix 1: consistent lock ordering

Always acquire multiple locks in the same global order across every code path — typically by sorting locks by some stable identity (e.g. object id, an assigned rank).

```
import threading

lock_a = threading.Lock()
lock_b = threading.Lock()

def thread_1():
    with lock_a:
        with lock_b:             # both threads always take lock_a THEN lock_b
            print("Thread 1 got both locks")

def thread_2():
    with lock_a:                 # same order as thread_1, no deadlock possible
        with lock_b:
            print("Thread 2 got both locks")
```

15.3 Fix 2: timeouts with backoff

```
import threading, time, random

lock_a = threading.Lock()
lock_b = threading.Lock()
```

```
def acquire_both(first, second, name):
    while True:
        first.acquire()
        if second.acquire(timeout=0.05):
            try:
                print(f"{name} got both locks")
                return
            finally:
                second.release()
                first.release()
        else:
            first.release()
            time.sleep(random.uniform(0.01, 0.05)) # back off and retry

threading.Thread(target=acquire_both, args=(lock_a, lock_b, "T1")).start()
threading.Thread(target=acquire_both, args=(lock_b, lock_a, "T2")).start()
```

15.4 Livelock and starvation

- **Livelock:** threads keep changing state in response to each other (e.g. both backing off and retrying in lockstep) but make no real progress — different from deadlock in that the threads are actively running, just uselessly. Randomized backoff (as above) reduces the chance of this.
- **Starvation:** a thread never gets access to a resource because other threads keep getting priority — e.g. a low-priority reader in a reader/writer scheme that's constantly pre-empted by writers. Fair queuing or bounded waiting policies address this; Python's primitives are FIFO-ish under CPython but this isn't a strict language guarantee.

***Pitfall:** There is no reliable, safe way to force-kill a hung thread in Python. Prevention (consistent lock ordering, timeouts, minimal nested locking) is the only real defense — design for it up front rather than debugging it after the fact.*

16. Threading vs Multiprocessing vs Asyncio

	threading	multiprocessing	asyncio
Best for	I/O-bound	CPU-bound	I/O-bound, very high concurrency
Parallelism	No (GIL-limited)	True (separate processes)	No (single thread, cooperative)
Memory	Shared	Isolated (needs IPC)	Shared
Overhead per unit	~KB, fast to spawn	~MB, slower to spawn	Very low (coroutines)
Failure isolation	Low (shared process)	High (separate process)	Low
Typical count	Tens to low hundreds	= CPU core count	Thousands+
Needs sync primitives	Yes (Lock, Queue...)	Yes (different set, via IPC)	Usually less (single-threaded)

These are complementary, not mutually exclusive — a common production pattern is asyncio for the main event loop with a `ThreadPoolExecutor` for the occasional blocking/legacy call (`loop.run_in_executor`), and multiprocessing (or a separate service) for genuinely CPU-heavy work like image processing or ML inference.

```
import asyncio
from concurrent.futures import ThreadPoolExecutor

def blocking_legacy_call(x):
    import time; time.sleep(1); return x * 2

async def main():
    loop = asyncio.get_running_loop()
    with ThreadPoolExecutor() as pool:
        result = await loop.run_in_executor(pool, blocking_legacy_call, 21)
        print(result)

asyncio.run(main())
```

17. Debugging & Observing Threads in Production

17.1 Enumerating and inspecting live threads

```
import threading

print("Active threads:", threading.active_count())
for t in threading.enumerate():
    print(f"  {t.name:20s} alive={t.is_alive()} daemon={t.daemon}")

print("Current thread:", threading.current_thread().name)
print("Main thread:", threading.main_thread().name)
```

17.2 Dumping stack traces of every thread (diagnosing a hang)

When a program appears frozen, dumping every thread's current stack frame is the fastest way to find which lock or I/O call is stuck.

```
import faulthandler, signal, sys

# Registers a handler: send SIGUSR1 to the process to print all thread stacks
# without stopping it (Unix only).
faulthandler.register(signal.SIGUSR1, file=sys.stderr)

# Or dump immediately, on demand:
faulthandler.dump_traceback()

# Cross-platform alternative using the traceback + sys._current_frames() API:
import sys, traceback

def dump_all_threads():
    frames = sys._current_frames()
    for thread_id, frame in frames.items():
        print(f"--- Thread {thread_id} ---")
        traceback.print_stack(frame)

dump_all_threads()
```

17.3 Logging best practices for multithreaded programs

```
import logging

logging.basicConfig(
    level=logging.DEBUG,
    format="%(asctime)s.%(msecs)03d [%(threadName)-15s] %(levelname)s %(message)s",
    datefmt="%H:%M:%S",
)

# The logging module is itself thread-safe: handlers use an internal lock,
# so interleaved log calls from multiple threads won't corrupt a single line.
```

17.4 Detecting a suspected deadlock in tests

```
import threading, time

def run_with_deadlock_guard(fn, timeout=5):
    result = {}
    def target():
```

```
        result["value"] = fn()
t = threading.Thread(target=target)
t.daemon = True
t.start()
t.join(timeout)
if t.is_alive():
    raise TimeoutError(f"{fn.__name__} appears deadlocked (exceeded {timeout}s)")
return result.get("value")
```

18. Design Patterns for Real Systems

18.1 Worker pool with graceful shutdown

```
import threading, queue, logging

logging.basicConfig(level=logging.INFO, format="[%s] %(message)s" % (threading.currentThread().getName(),))

class WorkerPool:
    def __init__(self, num_workers, handler):
        self.tasks = queue.Queue()
        self.handler = handler
        self._stop = threading.Event()
        self.workers = [
            threading.Thread(target=self._loop, name=f"Worker-{i}")
            for i in range(num_workers)
        ]

    def start(self):
        for w in self.workers:
            w.start()

    def submit(self, item):
        self.tasks.put(item)

    def _loop(self):
        while not self._stop.is_set() or not self.tasks.empty():
            try:
                item = self.tasks.get(timeout=0.2)
            except queue.Empty:
                continue
            try:
                self.handler(item)
            except Exception:
                logging.exception("Task failed")
            finally:
                self.tasks.task_done()

    def shutdown(self, wait=True):
        self._stop.set()
        if wait:
            for w in self.workers:
                w.join()

pool = WorkerPool(4, handler=lambda item: logging.info(f"processing {item}"))
pool.start()
for i in range(20):
    pool.submit(i)
pool.shutdown(wait=True)
```

18.2 Thread-safe singleton / lazy initialization

```
import threading

class Singleton:
    _instance = None
    _lock = threading.Lock()
```

```

def __new__(cls):
    if cls._instance is None:
        with cls._lock:
            # double-checked locking: re-check inside the lock in case
            # another thread already created it while we were waiting
            if cls._instance is None:
                cls._instance = super().__new__(cls)
    return cls._instance

```

18.3 Read-write lock (favoring concurrent readers)

The standard library has no built-in reader/writer lock, but one is easy to compose from a Lock and a counter when reads vastly outnumber writes.

```

import threading

class ReadWriteLock:
    def __init__(self):
        self._readers = 0
        self._readers_lock = threading.Lock()
        self._writer_lock = threading.Lock()

    def acquire_read(self):
        with self._readers_lock:
            self._readers += 1
            if self._readers == 1:
                self._writer_lock.acquire() # first reader blocks writers

    def release_read(self):
        with self._readers_lock:
            self._readers -= 1
            if self._readers == 0:
                self._writer_lock.release() # last reader unblocks writers

    def acquire_write(self):
        self._writer_lock.acquire()

    def release_write(self):
        self._writer_lock.release()

rw = ReadWriteLock()

def reader():
    rw.acquire_read()
    try:
        pass # read shared_data
    finally:
        rw.release_read()

def writer():
    rw.acquire_write()
    try:
        pass # mutate shared_data
    finally:
        rw.release_write()

```

18.4 Bounded, self-throttling producer

Combine a bounded Queue with a ThreadPoolExecutor to build a pipeline where producers automatically slow down when consumers fall behind, avoiding unbounded memory growth.

```
import queue, threading
from concurrent.futures import ThreadPoolExecutor

results_queue = queue.Queue(maxsize=50)  # backpressure: producers block once full

def produce(n):
    for i in range(n):
        results_queue.put(i)  # blocks automatically if consumers can't keep up
    results_queue.put(None)

def consume():
    while True:
        item = results_queue.get()
        if item is None:
            break
        # process item

threading.Thread(target=produce, args=(1000,)).start()
with ThreadPoolExecutor(max_workers=4) as pool:
    for _ in range(4):
        pool.submit(consume)
```

19. Best Practices Checklist

- **Default to `ThreadPoolExecutor`** for one-off concurrent work; drop to raw `Thread` only when you need long-lived, individually-managed workers.
- **Threading is for I/O-bound work.** For CPU-bound work, use multiprocessing, a GIL-releasing C extension, or a separate service.
- **Keep critical sections tiny.** Never hold a lock across I/O, sleeps, or expensive computation.
- **Prefer `with lock:`** over manual acquire/release so exceptions can't leak an unreleased lock.
- **Prefer `queue.Queue`** over hand-rolled Condition-based producer/consumer code wherever it fits — it's simpler and already correct.
- **Use `RLock` only when a thread may re-enter its own lock** (nested method calls); otherwise use a plain `Lock` so accidental re-entrancy is caught as a bug, not hidden.
- **Acquire multiple locks in a single, consistent global order everywhere** to eliminate the most common deadlock.
- **Name your threads** (`name=`) and configure `%(threadName)s` in logging — future-you debugging a hang at 2am will be grateful.
- **Use daemon threads only for disposable background work**; for anything with side effects, use an `Event` for graceful, cooperative shutdown.
- **Never assume you can kill a thread.** Design cancellation cooperatively from the start.
- **Always call `task_done()`** in a `finally` block when using `Queue.join()`.
- **Check `Future.result()` (or `.exception()`)** for every submitted task — silent exceptions in worker threads are a top source of "why did this just stop working" bugs.
- **Measure before optimizing.** Profile whether your bottleneck is actually I/O-bound before reaching for threads at all — sometimes a single async event loop or batched I/O beats a thread pool outright.

End of guide. Every code sample above is self-contained and runnable as-is with the Python standard library — no third-party dependencies required.